HOMEWORK 15

This is the last HW of this semester. You only need to submit the following Exercises from Artin's book and Problems 1, 2, 3 below. The rest part is for the summer break.

Exercises: 6.3, 6.6, 7.1, 7.2, 7.3, 7.4, 7.7, 7.8, 7.9, 8.3, 8.4, 8.5. pages 223-225,

1. Problems related to Ex 6.3, page 223

The following several problems are related to Ex.6.3, page 223.

Problem 1. *Let G be a group and $H < G$ be a subgroup of G. Let $C_G(H) = \{g \in G : gh = hg, \forall h \in H\}$. The group $C_G(H)$ is called the centralizer of H in G. Show that $C_G(H)$ is indeed a subgroup of G and $C_G(H)$ is a normal subgroup of $N_G(H)$. Moreover, show that there is natural injective homomorphism $\phi : N_G(H)/C_G(H) \to \mathrm{Aut}(H)$.*

Note that in general, H is not a subgroup of $C_G(H)$. When is H a subgroup of $C_G(H)$?

Problem 2. (1) *Determine the center of S_n.*

(2) *Determine the center of $\mathrm{GL}_n(F)$. Here F is a fixed field.*
(3) *Let P be the subgroup of $\mathrm{GL}_n(F)$ which consists of all permutation matrices. Consider $C(P) = \{g \in \mathrm{GL}_n(F) : gx = xg, \forall x \in P\}$, the centralizer of P in $\mathrm{GL}_n(F)$. Note that $C(P) < N(P)$, where $N(P)$ is the normalizer of P in $\mathrm{GL}_n(F)$. The group $C(P)$ is called the centralizer of P in $\mathrm{GL}_n(F)$. Describe $C(P)$ explicitly.*

Problem 3. *Let P be the subgroup of $\mathrm{GL}_n(F)$ which consists of all permutation matrices. Let $N(P)$ be the normalizer of P in $\mathrm{GL}_n(F)$.*

(1) *Let $\sigma \in S_n$ be a permutation and let m_k be the number of k-cycles in the cycle decompositions of σ. For example, if $\sigma = (123)(45)(67)(8)(9) \in S_9$, then $m_1 = m_2 = 2$, $m_3 = 1$ and $m_k = 0$ for other k. Note that $n = \sum_{k=1}^n km_k$. Let P_σ be the corresponding permutation matrix. Show that the characteristic polynomial of P_σ is $\chi_{P_\sigma} = \prod_{k=1}^n (x^k - 1)^{m_k}$. For example, in the above example $\sigma = (123)(45)(67)(8)(9) \in S_9$, we have $\chi_{P_\sigma} = (x^3 - 1)(x^2 - 1)^2(x - 1)^2$.*
(2) *Let $g \in \mathrm{GL}_n(F)$, show that there exists an element $\sigma \in S_n$ such that $gP_\sigma \in C(P)$, the centralizer of P in $\mathrm{GL}_n(F)$.*
(3) *Describe $N(P)$. Moreover, show that there is an isomorphism $N(P)/C(P) \cong S_n$.*

The last part is Exercise 6.3, page 223.

2. Outer automorphism of S_6

Problem 4. *What are the class equations of $\mathrm{PSL}_2(\mathbb{F}_5)$ and $\mathrm{PGL}_2(\mathbb{F}_5)$? Show that $\mathrm{PSL}_2(\mathbb{F}_5)$ is simple using its class equation.*

Problem 5. *Show that $\mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$ and $\mathrm{PGL}_2(\mathbb{F}_5) \cong S_5$.*

As in the proof that the icosahedral group is isomorphic to A_5, one needs to find a set with 5 elements such that $\mathrm{PSL}_2(\mathbb{F}_5)$ acts on it. Consider the Sylow 2 subgroups. This is Exercise 8.4, page 287. It is a fact that any faithful subgroup of S_6 of order 60 is isomorphic to A_5. It is known that $\mathrm{PSL}_2(\mathbb{F}_5)$ is simple. You can use this fact if you think it is useful.

Let $X = \mathbb{P}^1(\mathbb{F}_5)$. We know that $\mathrm{PSL}_2(\mathbb{F}_5)$ and $\mathrm{PGL}_2(\mathbb{F}_5)$ act on X in a natural way. See last HW. Thus we get a group homomorphism $f : \mathrm{PSL}_2(\mathbb{F}_5) \to S_6 \cong Perm(X)$ and $g : \mathrm{PGL}_2(\mathbb{F}_5) \to S_6 \cong Perm(X)$. From last HW, we know f is injective and its image is a transitive subgroup of S_6. Denote $H := \mathrm{Im}(f)$, which is a subgroup of S_6.

Problem 6. *Let Y be the set of all conjugates of H. Namely, $Y = \left\{gHg^{-1} \mid g \in S_6\right\}$.*

1. *Show that $N_{S_6}(H) = \mathrm{Im}(g) \cong S_5$ and conclude that $|Y| = 6$.*
2. *Consider the action of S_6 on Y by conjugation. Namely $S_6 \times Y \to Y$, $(g, xHx^{-1}) \mapsto gxHx^{-1}g^{-1}$. This gives a homomorphism $\Phi : S_6 \to Perm(Y) \cong S_6$. Show that Φ is an isomorphism.*
3. *Show that Φ is not an inner automorphism.*

It is a fact that $\mathrm{Aut}(S_6)$ is generated by inner automorphisms and Φ.

3. HW for summer break

There is no need to submit the rest part of the HW.

Problem 7. *Let F be a field and T be the subgroup of $G = \mathrm{GL}_n(F)$ which consists of all diagonal matrices. Determine $C_G(T), N_G(T)$ and show that there is an isomorphism $N_G(T)/C_G(T) \cong S_n$.*

The next one is an analogue of the above.

Problem 8. *Let F be a field and $G = \mathrm{Sp}_4(F)$ be the symplectic group. Let T be the subgroup of G consisting of all diagonal matrices. Compute $N_G(T)/C_G(T)$.*

4. Some problems related to direct product

Problem 9. *Let G be a group and N, H be two subgroups with N normal. Suppose that $G = N \rtimes H$. Show that there is an isomorphism $G/N \cong H$.*

Problem 10. *Let G be a group and N be a normal subgroup. Denote the quotient group G/N by H and denote the quotient map $G \to H = G/N$ by π. Suppose that there is a group homomorphism $s : H \to G$ such that $\pi \circ s = \mathrm{id}_H$. Show that s is injective and $G = N \rtimes s(H)$.*

Problem 11. *Let G_1, G_2 be two groups and we consider the direct product $G_1 \times G_2$. Let $p_1 : G_1 \times G_2 \to G_1$ be the projection map defined by $p_1(g_1, g_2) = g_1$. Similarly, let $p_2 : G_1 \times G_2$ be the map $p_2(g_1, g_2) = g_2$. Consider the triple $(G_1 \times G_2, p_1, p_2)$. Suppose that we are given another triple (H, f_1, f_2), where H is a group, $f_i : H \to G_i$ is a group homomorphism for $i = 1, 2$. Show that there is a unique group homomorphism $\phi : H \to G_1 \times G_2$ such that $f_i = p_i \circ \phi$ for $i = 1, 2$. In other words, we have the following commutative diagram*

$$\begin{tikzcd}
 & H \arrow[dl, "f_1"'] \arrow[d, dotted, "\phi"] \arrow[dr, "f_2"] & \\
G_1 & G_1 \times G_2 \arrow[l, "p_1"] \arrow[r, "p_2"'] & G_2
\end{tikzcd}$$

In other words, the direct product satisfies the same kind universal property as direct product of vector spaces, see section 2 of these notes. Since direct sum of finite number of vector spaces is the same as the direct product of finite number of vector spaces, one might ask if the same is true for groups. The answer is No.

Problem 12. *Let G_1, G_2 be two groups and we consider the direct product $G_1 \times G_2$. Let $\iota_1 : G_1 \to G_1 \times G_2$ be the map defined by $\iota_1(g_1) = (g_1, 1)$. Similarly, let $\iota_2 : G_2 \to G_1 \times G_2$ be the map $\iota_2(g_2) = (1, g_2)$. Consider the triple $(G_1 \times G_2, \iota_1, \iota_2)$. Suppose that we are given another triple (H, f_1, f_2), where H is a group, $f_i : G_i \to H$ is a group homomorphism for $i = 1, 2$. We ask if the following is true. Is there a unique group homomorphism $\phi : H \to G_1 \times G_2$ such that $f_i = \phi \circ \iota_i$ for $i = 1, 2$? In other words, is there a group homomorphism $\phi : H \to G$ such that the following diagram is commutative?*

$$\begin{tikzcd}
G_1 \arrow[r, "\iota_1"] \arrow[dr, "f_1"'] & G_1 \times G_2 \arrow[d, dotted, "\phi"] & G_2 \arrow[l, "\iota_2"'] \arrow[dl, "f_2"] \\
 & H &
\end{tikzcd}$$

The answer is No.

Now the natural question arises. Given two groups G_1, G_2, is there a triple (G, ι_1, ι_2) with group homomorphisms $\iota_i : G_i \to G$ such that it is universal in the above sense? Namely, for any other triple (H, f_1, f_2), where H is a group, $f_i : G_i \to H$ is a group homomorphism for $i = 1, 2$. Then there a unique group homomorphism $\phi : H \to G_1 \times G_2$ such that $f_i = \phi \circ \iota_i$ for $i = 1, 2$. In other words, there is a group homomorphism $\phi : H \to G_1 \times G_2$ such that the following diagram is commutative.

$$\begin{tikzcd}
G_1 \arrow[r, "\iota_1"] \arrow[dr, "f_1"'] & G \arrow[d, dotted, "\phi"] & G_2 \arrow[l, "\iota_2"'] \arrow[dl, "f_2"] \\
& H &
\end{tikzcd}$$

The triple (G, ι_1, ι_2) indeed exists. It is called the free product of G_1 and G_2 and it is usually denoted by $G_1 * G_2$. The free product is much more complicate than direct product. Try to think about what the free product $C_2 * C_2$ is. Here C_2 is the group with 2 elements. At least how many elements are there? Try to find an answer online.

There is also a notion called **free group**, which is defined as follows. Let S be any set. The free group over S is a pair $(\iota, G(S))$, where $G(S)$ is a group and $\iota : S \to G(S)$ is a map (between two sets) such that for any other pair (f, H), where H is a group and $f : S \to H$ is a map (between two sets), there is a unique group homomorphism $\phi : G(S) \to H$ such that the following diagram is commutative

$$\begin{tikzcd}
S \arrow[rr, "\iota"] \arrow[dr] & & G(S) \arrow[dl, dotted, "\phi"] \\
& H &
\end{tikzcd}$$

Free groups are covered in sections 7.9 and 7.10 of Artin's book. We don't have time to cover this. If you have time, read this part. If S is a singleton, (which means that $|S| = 1$), then $G(S) \cong \mathbb{Z}$. If $|S| > 1$, then $G(S)$ is not abelian anymore. Moreover, if $S = \{x_1, x_2\}$ contains 2 elements, then one can check that $G(S) = \mathbb{Z} * \mathbb{Z}$, which is the free product of two $\mathbb{Z}$, which is just the free group of a singleton. More precisely, if we write $S_1 = \{x_1\}, S_2 = \{S_2\}$, then $S = S_1 \coprod S_2$ (here $\coprod$ means disjoint union), then $G(S) = G(S_1) * G(S_2)$. This is true in general (namely without assuming that S_1, S_2 are singletons). These facts could be proved using the universal properties defined above. Try to check these facts without using the constructions of free groups and free products.

5. Problems on double cosets

The next several problems are about double coset decomposition. It could be in HW12. Since they are complicate, it is more appropriate for you to do them in the summer break.

Problem 13. *Let F be a field and let $W = F^2$, the two dimensional vector space over F. After choosing the standard basis $\mathcal{B} = \{\epsilon_1, \epsilon_2\}$, we can identify $\mathrm{GL}(W)$ with $\mathrm{GL}_2(F)$. Let $\mathcal{B}' = \{e, f\}$ be another basis of W, where $e = \epsilon_1 + \epsilon_2, f = \epsilon_1 - \epsilon_2$. Consider the subgroup*

$$H = \{g \in \mathrm{GL}(W) : ge \in \mathrm{Span}\{e\}, gf = f\}.$$

Compute the double cosets

$$H \backslash \mathrm{GL}(W)/B,$$

where $B \subset \mathrm{GL}(W)$ is the upper triangular subgroup. If we want to make it coordinate free, then

$$B = \{g \in \mathrm{GL}(W) : g\epsilon_1 \in \mathrm{Span}\{\epsilon_1\}\}.$$

There is a higher dimensional version of this problem.

Let $J_2 = \begin{bmatrix} & 1 \\ 1 & \end{bmatrix}$. Recall that the symplectic group $\mathrm{Sp}_4(F)$ is defined by

$$\mathrm{Sp}_4(F) = \left\{ g \in \mathrm{GL}_4(F) | g^t \begin{bmatrix} & J_2 \\ -J_2 & \end{bmatrix} g = \begin{bmatrix} & J_2 \\ -J_2 & \end{bmatrix} \right\}.$$

Problem 14. *Let B be the upper triangular subgroup of $\mathrm{Sp}_4(F)$. Compute the double coset*

$$B\backslash \mathrm{Sp}_4(F)/B.$$

Apparently, this problem has a higher dimensional version.

Problem 15. (1) *Let $G = \mathrm{GL}_n(\mathbb{R})$ and $K = \mathrm{O}_n(\mathbb{R})$. Describe the double coset $K\backslash G/K$.*
(2) *Let $G = \mathrm{GL}_n(\mathbb{C})$ and $K = \mathrm{U}_n(\mathbb{C})$. Describe the double coset $K\backslash G/K$.*

This is related to the Cartan decomposition. See HW 8.

6. Projective space and projective linear group

In some sense, group theory is useful because of group actions. There are many natural group actions and one very useful fact is the bijection $G/\mathrm{Stab}(x) \cong O_x$. Recall that one important application of these ideas is to prove the 3 Sylow Theorems. In order to use this, one needs to keep in mind several natural group actions. For example, given a group G and a subgroup H, we have the following natural group actions. (1) The group G acts on itself by left multiplication. (2) The group G acts on G by conjugation. (3) The group G acts on the left coset space G/H by left multiplication. (4) The group G acts on the power set $\mathcal{P}(G)$ by left multiplication, where the power set $\mathcal{P}(G)$ is the set of all subsets of G. (5) The group G acts on $\mathcal{P}(G, k)$ by left multiplication, where $\mathcal{P}(G, k)$ is the subset of $\mathcal{P}(G)$ which is consisting of order k subsets of G. (6) The group $\mathrm{GL}_n(F)$ acts on F^n, where F is a field. (7) The group D_{2n} acts on a regular n-polygon. (8) The tetrahedral group T acts on a tetrahedron. (9) The octahedral group acts on a cube or an octahedron. (10) The icosahedral group acts on a dodecahedron or an icosahedron. In the following, we give another very useful group action.

There following is a review of projective space, which has been appeared in a previous homework.

Let F be a field and n be a positive integer. We introduce an equivalence relation on F^{n+1} as follows. For $\alpha, \beta \in F^{n+1}$, we say that $\alpha \sim \beta$ if there is an element $c \in F^\times$ such that $\beta = c\alpha$.

Problem 16. *Show that $R = \{(\alpha, \beta) \in F^{n+1} \times F^{n+1} : \alpha \sim \beta\}$ defines an equivalence relation on F^{n+1}.*

Denote $\mathbb{P}^n(F) := F^{n+1}/\sim$, which is the set of equivalence classes of the above equivalence relation. For example, $\mathbb{P}^1(F)$ is just the set of all lines which passes the origin. The set $\mathbb{P}^n(F)$ is called the n-dimensional projective space. Here we only view it as a set even it has more structures.

An element of $\mathbb{P}^n(F)$ is of the form $[\alpha]$, where $\alpha \in F^{n+1}$ and $[\alpha]$ denotes the equivalence class of α. Note that $[\alpha] = [\beta]$ if and only if $\beta = c\alpha$ for some $c \in F^\times$.

There is a group action of $\mathrm{GL}_{n+1}(F)$ on $\mathbb{P}^n(F)$ defined by

$$g.[\alpha] = [g\alpha], g \in \mathrm{GL}_{n+1}(F), \alpha \in F^{n+1}.$$

Consider the group $\mathrm{PGL}_{n+1}(F) = \mathrm{GL}_{n+1}(F)/Z$, where $Z = \{aI_{n+1}, a \in F^\times\}$ is the center of $\mathrm{GL}_{n+1}(F)$. The group $\mathrm{PGL}_{n+1}(F)$ is called the **projective general linear group** over F. An element of $\mathrm{PGL}_{n+1}(F)$ is of the form gZ, where $g \in \mathrm{GL}_{n+1}(F)$ and $gZ = hZ$ if and only if $g^{-1}h \in Z$, or $g = hz$ for some $z \in Z$. Similarly, we consider the group $\mathrm{PSL}_{n+1}(F) = \mathrm{SL}_{n+1}(F)/Z_0$, where $Z_0 = \{aI_{n+1} : a \in F^\times, a^{n+1} = 1\}$ is the center of $\mathrm{SL}_{n+1}(F)$.

Problem 17. *Show that the map $\mathrm{PGL}_{n+1}(F) \times \mathbb{P}^n(F) \to \mathbb{P}^n(F)$ defined by*

$$(gZ, [\alpha]) \mapsto [g\alpha]$$

is well-defined and it defines a group action of $\mathrm{PGL}_{n+1}(F)$ on $\mathbb{P}^n(F)$. Similarly, show that there is a group action of $\mathrm{PSL}_{n+1}(F)$ on $\mathbb{P}^n(F)$.

Problem 18. *Let F be any field and let B be the upper triangular subgroup of $\mathrm{GL}_2(F)$. Using the group action of $\mathrm{GL}_2(F)$ on $\mathbb{P}^1(F)$, show that there is a bijective map*

$$\mathrm{GL}_2(F)/B \cong \mathbb{P}^1(F).$$

Conclude that $B\backslash \mathrm{GL}_2(F)/B$ is consisting of two elements.

Hint for the second part. The double coset $B\backslash \mathrm{GL}_2(F)/B$ can be viewed as the orbit space of the natural action of B on the space $\mathrm{GL}_2(F)/B$, which can be realized on the natural action of B on $\mathbb{P}^1(F)$ using the above bijection.

This double coset $B\backslash \mathrm{GL}_2(F)/B$ was computed in previous HWs, which we proved using elementary row operations from linear algebra. Here we give a group theoretic solution to this for the small group $\mathrm{GL}_2(F)$. This problem could be generalized to more general situations but it needs some terminologies that we lack at this moment. Anyway, the following is a similar situation.

Problem 19. *Let $G = \mathrm{GL}_{n+1}(F)$. Let $P_{n,1}$ be the subgroup of G*

$$P_{n,1} = \left\{ g = \begin{pmatrix} g_1 & * \\ & a \end{pmatrix} \mid g \in \mathrm{GL}_n(F), a \in F^\times \right\}.$$

Show that there is a bijective map

$$\mathrm{GL}_{n+1}(F)/P_{n,1} \cong \mathbb{P}^n(F).$$

What is $P_{n,1}\backslash \mathrm{GL}_{n+1}/P_{n,1}$? What is $B\backslash \mathrm{GL}_{n+1}/P_{n,1}$?

7. Upper half plane

Let $i = \sqrt{-1}$. We consider the upper half plane $\mathcal{H} = \{x + yi|\ x, y \in \mathbb{R}, y > 0\}$ of the complex plane.

Problem 20. *Consider the map $\mathrm{SL}_2(\mathbb{R}) \times \mathcal{H} \to \mathcal{H}$ defined by*

$$\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}, z \right) \mapsto \frac{az+b}{cz+d}, \qquad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{R}), z \in \mathcal{H}.$$

1. *Show that the above map is well-defined and indeed defines a group action of $\mathrm{SL}_2(\mathbb{R})$ on $\mathcal{H}$. Moreover, show that the center of $\mathrm{SL}_2(\mathbb{R})$ acts trivially on $\mathcal{H}$ and thus it defines a group action $\mathrm{PSL}_2(\mathbb{R})$ on $\mathcal{H}$.*
2. *Find the stabilizer of $i \in \mathcal{H}$, namely, compute the group $\mathrm{Stab}(i) = \{g \in \mathrm{SL}_2(\mathbb{R}) | gi = i\}$.*
3. *Find the orbit O_i of i. Explicate the bijection $\mathrm{SL}_2(\mathbb{R})/(\mathrm{Stab}(i)) \cong O_i$.*

The above action can also be explained using the action of $\mathrm{PSL}_2(\mathbb{R})$ on $\mathbb{P}^1(\mathbb{C})$. Explain it.
This example of group action is important in number theory.

8. Orthogonal groups

The next problem was covered in class several weeks ago.

Problem 21. *Let $n > 1$ be a positive integer and consider the group*

$$\mathrm{SO}_n(\mathbb{R}) = \left\{ g \in \mathrm{GL}_n(\mathbb{R}) : gg^t = I_n, \det(g) = 1 \right\}.$$

Consider the subgroup H of $\mathrm{SO}_n(\mathbb{R})$ defined by

$$H = \left\{ \begin{bmatrix} h & \\ & 1 \end{bmatrix}, h \in \mathrm{SO}_{n-1}(\mathbb{R}) \right\}.$$

Show that there is a bijection

$$G/H \cong S^{n-1},$$

where $S^{n-1} = \left\{ (x_1, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 = 1 \right\}$, which is the standard $(n-1)$-sphere. Similarly, we consider the group $\mathrm{SU}_n = \{g \in \mathrm{GL}_n(\mathbb{C}) : gg^* = I_n, \det(g) = 1\}$. We view SU_{n-1} as a subgroup of SU_n via the embedding

$$h \mapsto \begin{bmatrix} h & \\ & 1 \end{bmatrix}, h \in \mathrm{SU}_{n-1}.$$

Show that there is a bijection

$$\mathrm{SU}_n/\mathrm{SU}_{n-1} \cong S^{2n-1}.$$

Problem 22. *Let $p > 2$ be a prime number and n be a positive integer. Consider the group*

$$\mathrm{SO}_n(\mathbb{F}_p) = \left\{ g \in \mathrm{GL}_n(\mathbb{F}_p) |\ gg^t = I_n, \det(g) = 1 \right\}.$$

Compute the order of $\mathrm{SO}_n(\mathbb{F}_p)$.

Hint: If you use the method the last problem, you need to know the order of the sets
$$X_n := \left\{(x_1, \dots, x_n) \in \mathbb{F}_p^n : x_1^2 + x_2^2 + \cdots + x_n^2 = 1.\right\}$$
It is not easy to compute this. The answer is
$$|X_n| = \begin{cases} p^{n-1} + (-1)^{\frac{n-1}{2}\cdot\frac{p-1}{2}} p^{\frac{n-1}{2}} & 2 \nmid n \\ p^{n-1} - (-1)^{\frac{n}{2}\cdot\frac{p-1}{2}} p^{\frac{n}{2}-1} & 2 | n \end{cases}$$
This is Proposition 8.6.1, page 102 of Ireland-Rosen: A classical introduction to modern number theory (2nd edition). See also this link.

The group $\mathrm{SO}_n(\mathbb{F}_p)$ is still the group which preserve a symmetric bilinear form on vector spaces over $\mathbb{F}_p$. But this time, this bilinear form is not an inner product. Inner products are only defined on vector spaces over $\mathbb{R}$ or $\mathbb{C}$, while bilinear forms can be defined over any fields. There is also a way to define $\mathrm{U}_n(\mathbb{F}_p)$ and $\mathrm{SU}_n(\mathbb{F}_p)$ and similarly one can ask how many elements are there in these groups.

The orthogonal groups over finite fields also depends on the symmetric bilinear form defined on that. Classifying symmetric bilinear forms over $\mathbb{F}_p$ is also an interesting question. We give one example below. Consider the group
$$\mathrm{SO}_{1,1}(\mathbb{F}_p) = \left\{ g \in \mathrm{GL}_2(\mathbb{F}_p) : g \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} g^t = \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \right\}.$$

Problem 23. *Let $p > 2$ be a prime.*

(1) *Show that* $|\mathrm{SO}_{1,1}(\mathbb{F}_p)| = p - 1$.
(2) *Show that* $|\mathrm{SO}_2(\mathbb{F}_p)| = p - (-1)^{\frac{p-1}{2}}$.
(3) *In particular, if* $p \equiv 1 \bmod 4$ *(for example, if* $p = 5, 13, 17 ...$*), then* $|\mathrm{SO}_{1,1}(\mathbb{F}_p)| \cong |\mathrm{SO}_2(\mathbb{F}_p)|$. *Is it true that* $\mathrm{SO}_2(\mathbb{F}_p) \cong \mathrm{SO}_{1,1}(\mathbb{F}_p)$ *if* $p \equiv 1 \bmod 4$*? If so, prove it.*

Try to generalize the last part to general n by considering the corresponding symmetric bilinear forms. Hint: Question: What is special for p with $p \equiv 1 \bmod 4$? Answer: the equation $x^2 + 1 = 0$ has a solution in $\mathbb{F}_p$.